



The Franz Josef Glacier

One of two glaciers that descend steeply from the Southern Alps toward the west coast of New Zealand's South Island

Named in 1865 after the Emperor of Austria, the Franz Josef Glacier is about 7 miles (12km) long. Its lower parts used to flow through a temperate rain forest near the coast. However, since 2008 its terminus has retreated rapidly and now hardly reaches the tree line.

The glacier is known to advance and retreat cyclically according to changes in regional climate. It is currently shrinking, but between 1983 and 2008 it advanced 1 mile (1.5 km). The driving force behind the glacier's whole existence is a high rate of precipitation in the upper area of the Southern Alps—up to 100 ft (30m) of snow falls onto the glacier's ice accumulation area each year—and annual changes in this precipitation rate can affect the glacier's behavior for years to come.



△ AQUA TUNNEL

One part of the glacier contains tunnels carved out by meltwater. This one, dubbed the Aqua Tunnel, is large enough to walk through.



△ BROKEN FLOES

The surface of the glacier's meltwater lake sometimes freezes over in winter. The floes (flat expanses of ice) later break up into angular slabs.

▷ MELTWATER LAKE

At the foot of the glacier stands a 4-mile- (7-km-) long meltwater lake. A century ago, the valley was still filled with the glacier itself.

